A travel agency can warn its clients two hours before a flight, if the data shows which flights will go wrong

TMS · NYC flights 2023 · 435,352 flights leaving JFK, LaGuardia and Newark [01]

Q1. Which flights will leave more than 15 minutes late or be cancelled, so the agency can warn clients two hours before departure?

Q2. Which airlines, airports, hours and months are the most and least reliable, and how much does the weather explain?

Q3. Which flights will be cancelled, so clients can be rebooked before it happens?

Data: nycflights23 (C. Ismay, S. P. Couch, H. Wickham, 2025, CC0). Training: January to September 2023. Test: October to December 2023, used once.

Five files, 435,352 flights, and two traps that would have fooled any model

Problem found	Rows	Decision
Cancelled flights (no departure time)	10,738	keep, they are the target
Tail number missing	1,913	keep, never a feature
Tail not in planes file	9,371	keep, fill later
Departed, arrival data missing	715	keep, flag
Delays over 12 hours	365	keep, flag (real)
Weather hours missing	73	leave empty
Duplicates	0	none

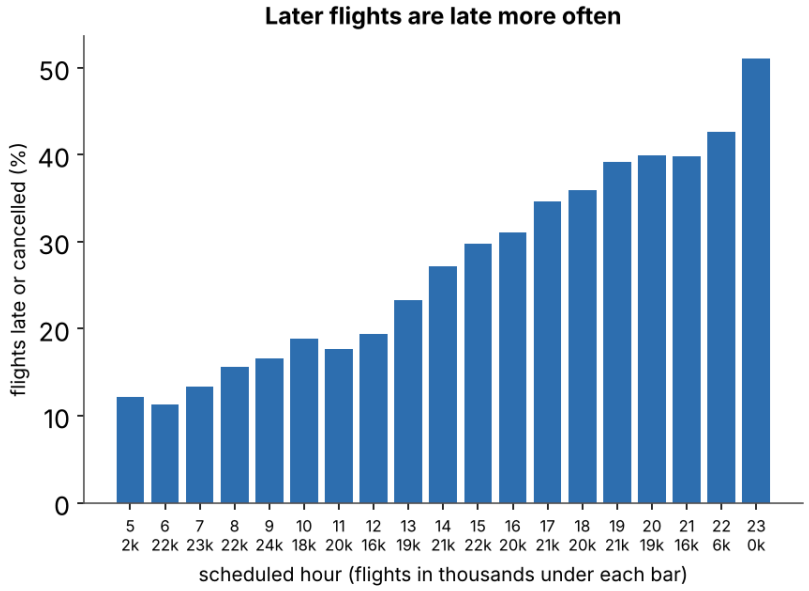
Every problem has a count, a decision and a reason; no flight was removed. [01]

Trap 1: the weather file is in UTC, the flights in New York time. A plain join pairs a flight with weather 4 to 5 hours off, so flights are converted to UTC first. [01]

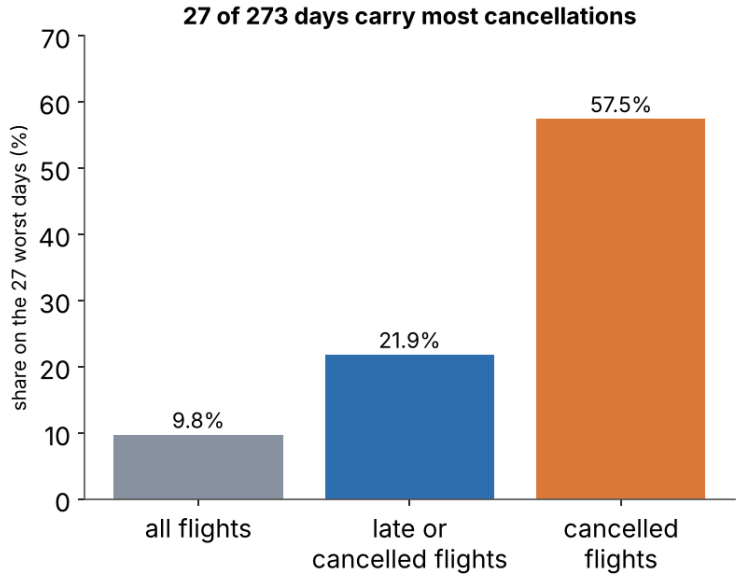
Trap 2: a missing tail number means cancelled. All 1,913 flights with no tail are cancelled flights, so a model would learn the answer from the gap. Plane data is never left empty. [01]

Source: notebook 01, section 6 (data report)

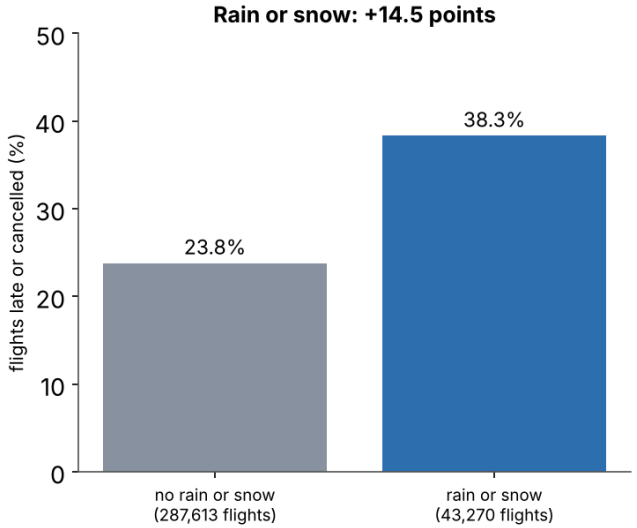
Findings 1 to 3: delays build up through the day, a few bad days carry the cancellations, and rain adds 14.5 points



13.4% of flights at 5-8 h are late or cancelled, **39.9%** at 19-22 h: gap **26.5 points** (interval 24.2 to 28.9). [02]
Action: warn evening flights first.



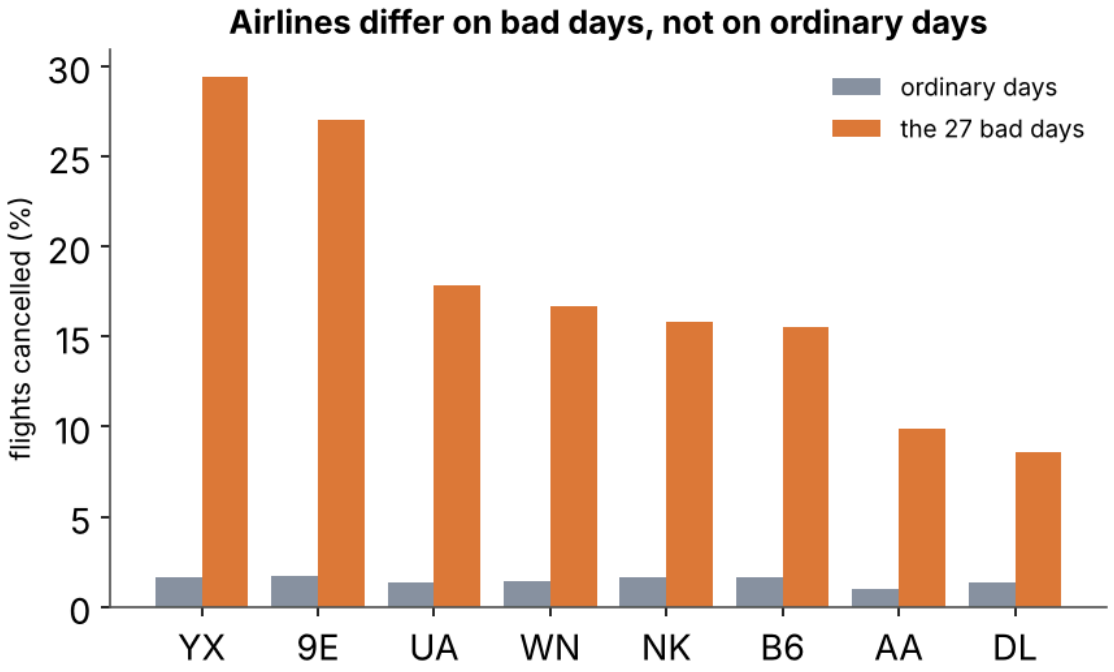
27 days with 45% or more flights late or cancelled hold **57.5%** of all cancellations (27 random days: 10%). [02]
Action: needs a real-time bad-day alert.



Gap **+14.5 points** (interval 11.1 to 18.1); still 1.48 times the expected rate after hour, month and airport. [02]
Action: use weather in the model.

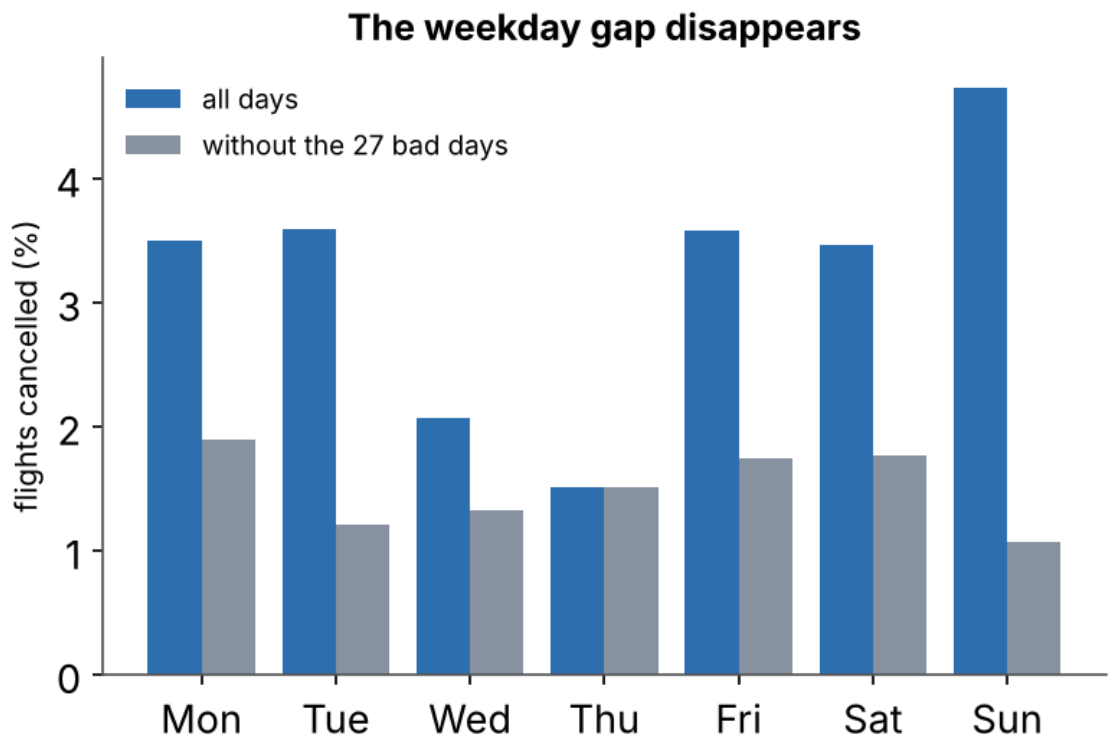
Source: notebook 02, training months (330,883 flights); intervals resample whole days

Findings 4 and 5: airlines differ on bad days, and the 'safe Thursday' was only luck



Finding 4. On ordinary days big airlines cancel 1.0% to 1.7% of flights; on bad days Republic (YX) cancels **29%** and Endeavor (9E) **27%**, Delta 9%. [02]

Action: rebook YX and 9E clients first on a bad day.



Finding 5 (a strange pattern). Thursday looked safest (1.5%) and Sunday worst (4.7%). Without the 27 bad days the gap nearly vanishes (Cramer's V 0.057 to 0.023). Thursday had 0 bad days, Sunday 6; a weekday with none happens in 8.7% of random placements. [02]

Action: do not tell clients Thursday is safer.

Source: notebook 02, training months

Destination airports fall into three groups, and each needs a different action

Group	Airports	Flights	Typical (Jan-Sep)	What the agency does
G1 short-haul, mild delays, most cancellations	57	101,287	late 21.2%, cancelled 4.3%, 588 miles; 81% YX and 9E	keep rebooking ready first
G2 main long-haul, delays build up through the day	45	226,896	late 27.8%; morning 15% to evening 42%, 1,426 miles	warn earlier on afternoon and evening flights, check connections
G3 chronic delays on few flights (BQN, SMF, ONT, PSE, ABQ, SJC, RNO)	7	2,374	late 53.1%, mean delay 41.5 minutes; 77% from JFK, 77% JetBlue	always warn and check connections; re-check monthly

9 destinations with fewer than 100 flights are labelled 'small'. k = 3 groups of destination airports (k-means on 8 delay columns). Honest limit: the silhouette score is 0.322, a weak structure; the same groups came back in 10 runs (adjusted Rand index at least 0.965). [03]

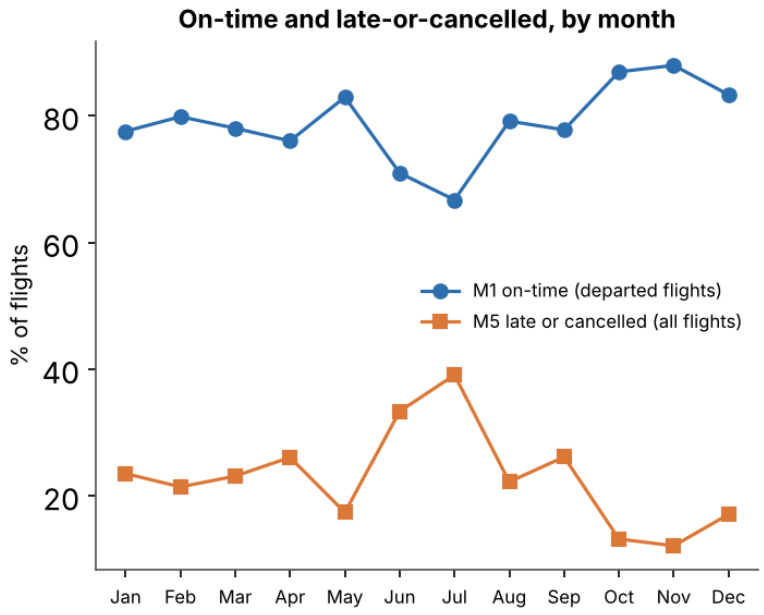
Groups are fitted on January to September only, so they can be used as a model column without leakage.

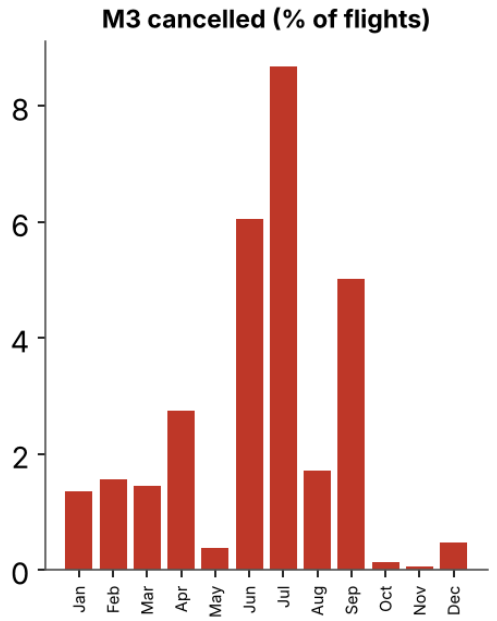
Source: notebook 03, sections 4 and 7

Five metrics tell the agency how reliable its flights are; July was the worst month

- **M1 on-time**: departed flights with delay of 15 minutes or less / departed flights (whole year 79.03%)
- **M2 average delay**: mean departure delay of departed flights, minutes (13.84)
- **M3 cancelled**: flights with no departure / all scheduled flights (2.47%)
- **M4 over an hour late**: delay above 60 minutes / departed flights (8.41%)
- **M5 late or cancelled**: delay above 15 minutes or no departure / all scheduled flights (22.91%)

Whole-year values [04]

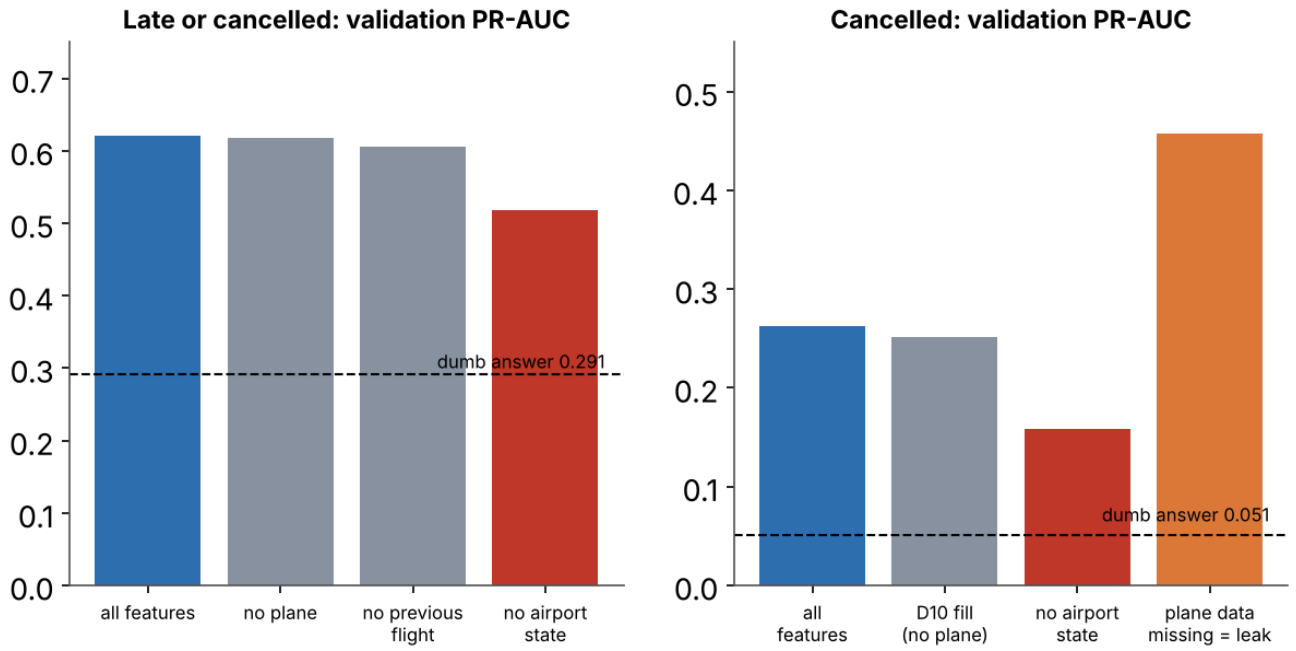




Unusual months. Only July's cancellation rate (8.68% of flights) is a statistical outlier (modified z-score 4.1 against a limit of 3.5). June and July were not wetter or windier; the 27 bad days hold 76% and 69% of their cancellations, and Newark and United stand out. Cause of the bad days: not explained by the available data. [04]

Source: notebook 04, sections 1, 2 and 4; flights = 435,352 (monthly 33,362 to 39,514)

The strongest new column is how late the airport already is, and no column uses information from after the prediction moment



Quick model trained on January-June and scored on July-September; dashed line = dumb answer. [05]

- **51 new columns**: calendar, route, plane, weather (two sets), airport traffic, **airport state** (delays of flights that left the same airport in the last 2 hours), previous flight of the same plane, airline and route history.
- **Rule:** prediction moment = scheduled departure minus 2 hours; only what is known then is used.
- **Leak tests pass**: no forbidden column; 1,000 flights recomputed by a slow loop agree; deleting all later flights changes nothing; strict weather is never later than the moment. It also caught a tie bug. [05]
- A naive empty plane column lifts cancellation PR-AUC to 0.458 (a leak); the random same-airline plane fill gives 0.263.

Source: notebook 05, sections 11 and 12

Both predictions are tested the honest way: time split, a dumb answer to beat, and no accuracy



Models are tuned on Jul-Sep (fit on Jan-Jun), then refit on Jan-Sep (330,883 flights) and scored once on Oct-Dec (104,469 flights). [06]

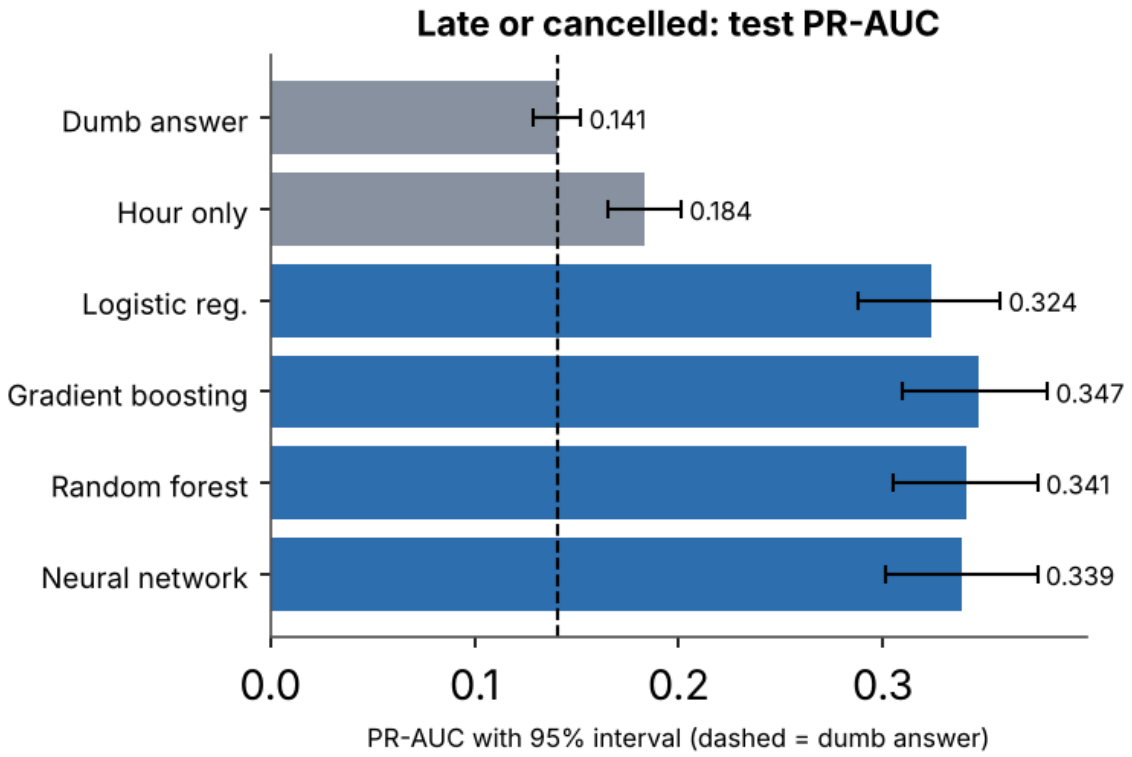
- **Dumb answer:** 'nobody is late' scores PR-AUC = the share of late flights: 0.1406 on test (cancelled: 0.0021). It is 85.9% accurate and flags nobody, so accuracy is never used. [06]
- **Hour-only rule** is a second bar to beat (late 0.1836). For cancellations it does not beat the dumb answer.
- **Test late rate is 14.1% against 25.7% in training** (drift), so each score is compared with the dumb answer of the same part.

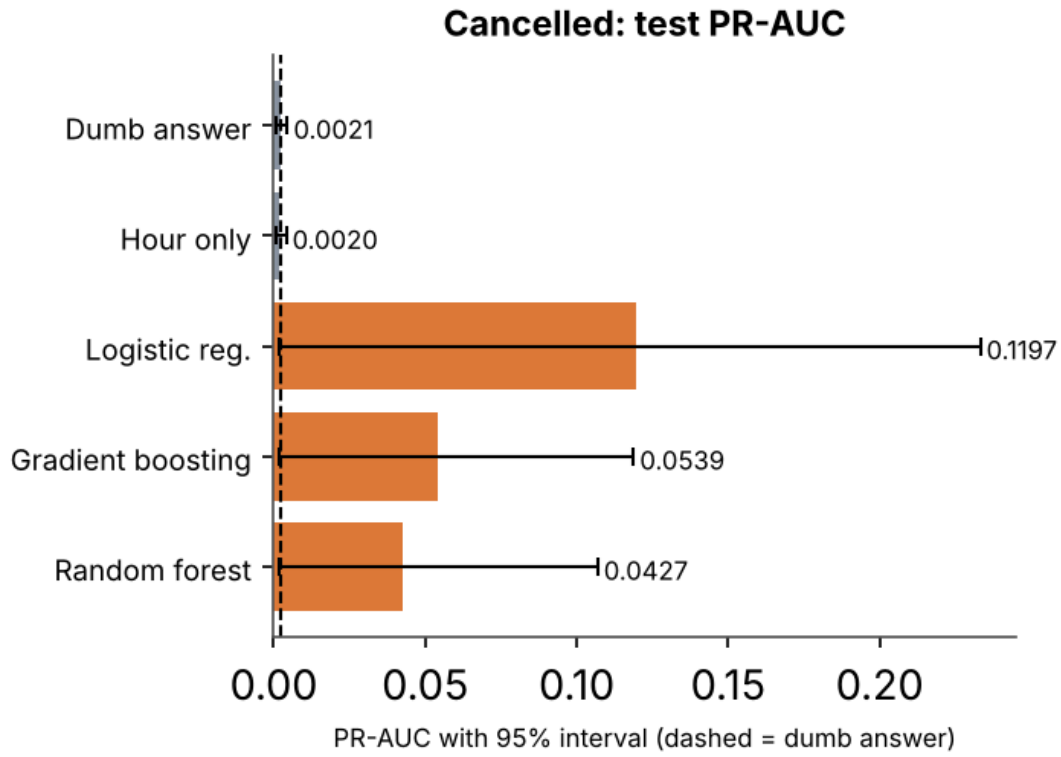
Columns removed (leakage list):

- departure and arrival times, delays, air time: written after the event
- tail number and 'tail missing': missing = cancelled for 1,913 flights
- flags and targets built from the outcome; flight number; year
- anything computed from flights after the prediction moment [06]

Source: notebook 06, sections A2, A5, B5

Every model beats the dumb answer, but the models cannot be told apart and cancellations stay hard





Late or cancelled (104,469 test flights): neural network 0.339, gradient boosting 0.347, random forest 0.341, logistic regression 0.324; all inside the same interval, about 2.4 times the dumb answer. Weather join choice changes the score by 0.004 or less (inside the interval). [06]

Cancelled (224 flights): logistic regression 0.120 against 0.0021, but the interval is wide (0.002 to 0.233): 115 of 224 cancellations are on one day, 18 December. Without week 51: 0.0071 against 0.0011. [06]

Weak spots: morning recall 5.8%, Newark 20.5% (late); all 39 Newark cancellations missed. The model flags Alaska, American and Endeavor less than their real risk. [06]

Source: notebook 06, sections E2, E3, D2, D3

Card 1: flag 11.5% of flights and catch 30.5% of late or cancelled flights, 2.7 times better than random

0.597
cut-off on the network score, chosen on Jul-Sep [07]

37.4%
precision, interval 34.2 to 40.6 (4,481 of 11,974 flags) [07]

30.5%
recall, interval 25.4 to 35.1 (of 14,685 cases) [07]

2.7x
better than picking 11,974 flights at random (14.1%) [07]

- **Question:** will the flight leave over 15 minutes late or be cancelled, two hours before departure?
- **Action:** message the client of each flagged flight and check connections; warn earlier on afternoon and evening flights (long-haul group G2).
- **Workload:** 130 flights a day in Oct-Dec.
- **Model:** neural network, 37 columns, trained Jan-Sep. PR-AUC 0.339 [0.301, 0.376] against dumb 0.141.
- **Weak spots:** mornings recall 5.8% (evenings 51.0%); Newark 20.5% (JFK 40.2%).
- **Re-check when** weekly recall is below 14% two weeks running (14 test weeks ranged 14.3% to 51.7%).
- **Human check:** confirm the flight status on the airline's feed; never auto-rebook; message says 'possible delay'.
- **Finding:** how late the airport already is beats weather ten times over (0.228 against 0.023).

Source: notebook 07, section 7 (card 1); flights = 104,469 test flights

Card 2: 5 flagged flights a day catch 42% of cancellations, but only 19 in 100 flags are real

0.137
cut-off on the probability, chosen on Jul-Sep [07]

19.4%
precision, interval 0.6 to 25.9 (94 of 484 flags) [07]

42.0%
recall, interval 1.0 to 60.5 (of 224 cancellations) [07]

91x
better than random (0.21% of flights) [07]

- **Question:** will the flight be cancelled, two hours before departure?
- **Action:** offer a rebooking to the client of each flagged flight; never rebook automatically. Strongest for short-haul G1 (precision 25.1%, 81 cases).
- **Workload:** 5.3 flights a day; top 1% of scores = 11 a day: precision 10.2%, recall 47.8%.
- **Model:** logistic regression, 37 columns. PR-AUC 0.120 [0.002, 0.233] against dumb 0.0021.
- **Weak spots:** none of 39 Newark cancellations found; recall 0% in October, 5% in November, 60% in December.
- **Re-check when** a bad-day block of 4 weeks (at least 30 cancellations) has recall under 23%. Weekly counts are too small.
- **Human check:** confirm the flight is still scheduled; a flag means 'high risk'.
- **Finding:** 115 of 224 cancellations fell on one day, 18 December (9.8% of that day's 1,174 flights; Endeavor and Republic). Weather was ordinary: cause not explained by the available data.

Precision is low in absolute terms and the result rests on one day; it is a ranking aid for rebooking capacity, not a promise.

Source: notebook 07, section 7 (card 2); flights = 104,469 test flights

AI helped to write and check the code; every number was re-computed and the limits are stated

Where AI helped

- drafting notebook code and the helper file
- debugging (memory crash, a tie bug in the previous-flight column, a divide-by-zero in the bootstrap)
- wording of explanations and of this deck

What was checked

- all counts of the data report match the plan; the 15 metric check values are asserted in code
- four leak tests pass; test part scored once; notebooks run top to bottom with identical numbers on a re-run
- every slide number has a line in the README ledger

Limits

- One year, three airports; weather is observed, not a forecast
- Train and test differ strongly: late 25.7% against 14.1%, cancelled 3.18% against 0.21%
- Cancellation precision is low, and half of the test cancellations are on one day
- Group structure is weak (silhouette 0.322)
- Models of the same family are not separable in the test interval

Next: a real weather forecast feed and a cancellation timestamp.

Source: README.md (how AI was used), notebooks 01 to 07